

Topic 1: Problem solving

Students are expected to develop a set of computational thinking skills that enable them to understand how computer systems work, and design, implement and analyse algorithms for solving problems.

Students should be given repeated opportunities to tackle computational problems of various sorts, including some substantial problem-solving tasks.

Students will be expected to use the pseudocode listed in *Appendix 5: Pseudocode command set* and the flowchart symbols shown in *Appendix 6: Flowchart symbols*.

	Students should:
1.1 Algorithms	1.1.1 Understand what an algorithm is, what algorithms are used for and be able to interpret algorithms (flowcharts, pseudocode, written descriptions, program code). 1.1.2 Understand how to create an algorithm to solve a particular problem, making use of programming constructs (sequence, selection, iteration) and using appropriate conventions (flowchart, pseudocode, written description, draft program code). 1.1.3 Understand the purpose of a given algorithm and how an algorithm works. 1.1.4 Understand how to determine the correct output of an algorithm for a given set of data. 1.1.5 Understand how to identify and correct errors in algorithms, including using trace tables. 1.1.6 Understand how to code an algorithm in a high-level language. 1.1.7 Understand how the choice of algorithm is influenced by the data structures and data values that need to be manipulated. 1.1.8 Understand how standard algorithms work (bubble sort, merge sort, linear search, binary search). 1.1.9 Be able to evaluate the fitness for purpose of algorithms in meeting specified requirements efficiently, using logical reasoning and test data.